BREADTH-FIRST SEARCH

LECTURE BY MIN WANG

LEARNING OBJECTIVES

- Graph Searching Context
- Breadth-First Search Strategy
- Breadth-First Search Algorithm

REQUIREMENTS

- You should have some basic understanding of Graphs representations.
- Graph G=(V, E)
- V=set of vertices
- E=set of edges
- If E contains edges like e={s, w}, unordered pairs, then It is a undirected graph
- If E contains edges like e=(s, w), ordered pairs, then it is a directed graph

GRAPH SEARCHING CONTEXT

- Graph

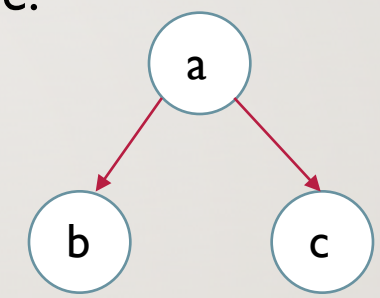
  ↓ traversal

- Tree

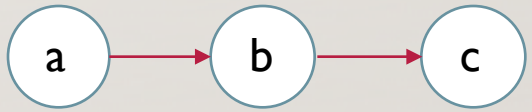
  ↓ traversal

- Sequence

One tree:



One traversal sequence:



BREADTH-FIRST SEARCH ALGORITHM STRATEGY

- Firstly, visit the vertex S and color it in Black

BREADTH-FIRST SEARCH ALGORITHM STRATEGY

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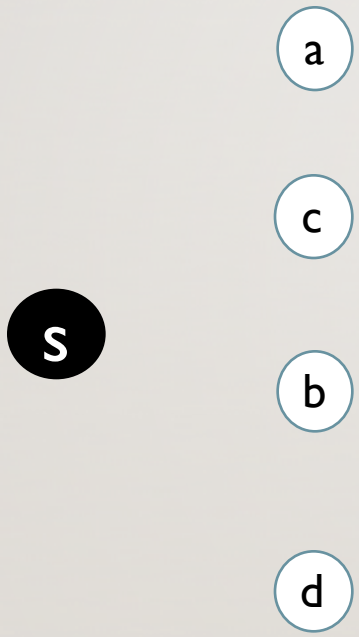

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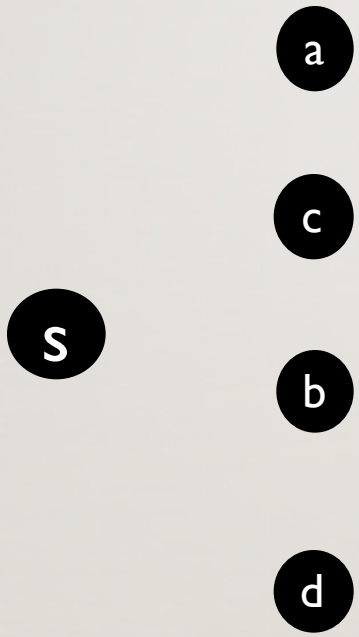

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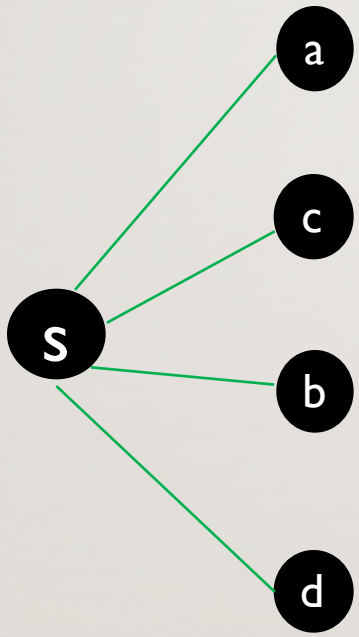

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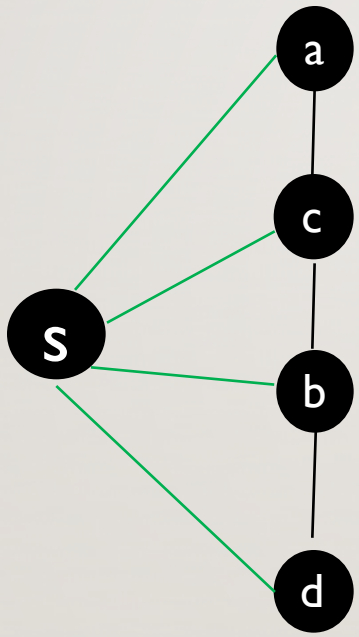

- There might have edges between each other among the list of neighbor vertices of vertex S. But they are not visited in the traversal
- Observe that the visited vertices and edges form an tree

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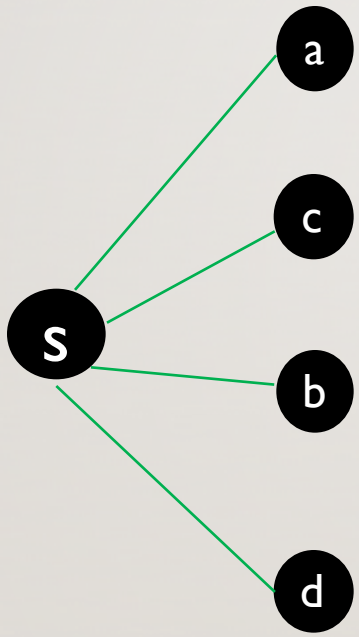

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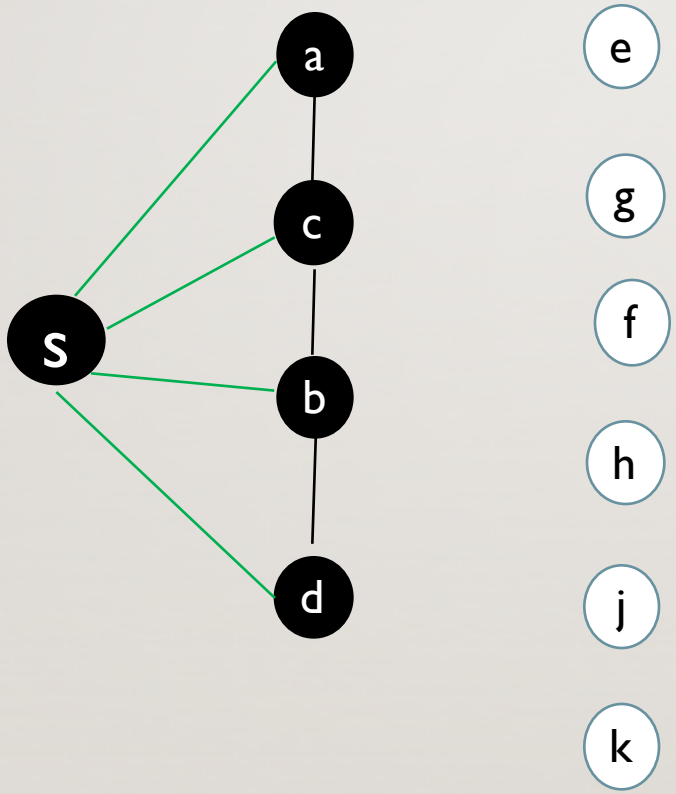

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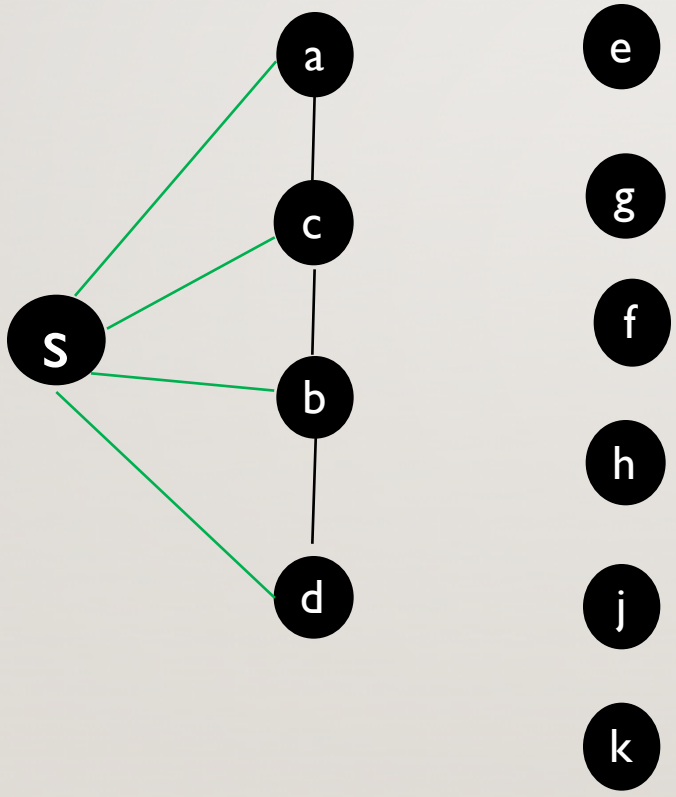

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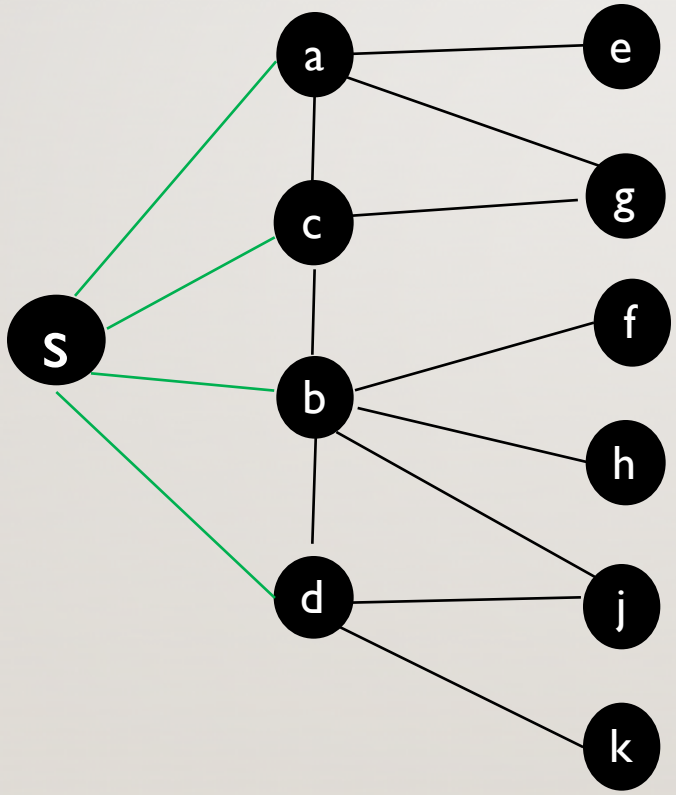

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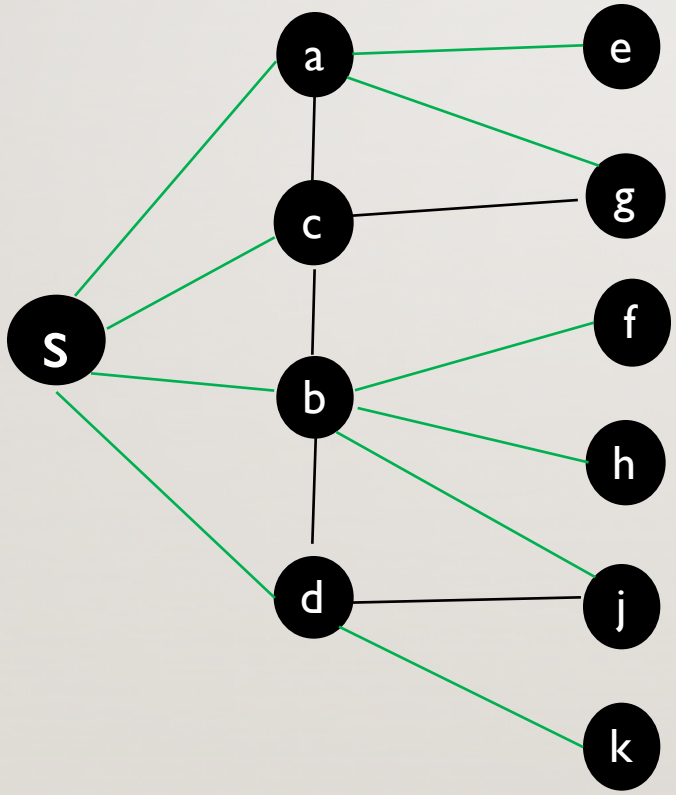

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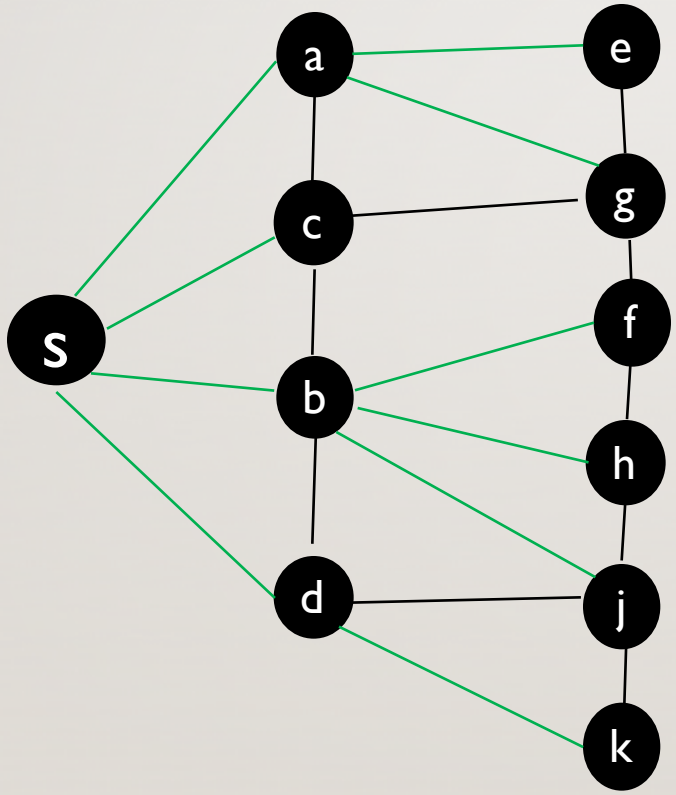

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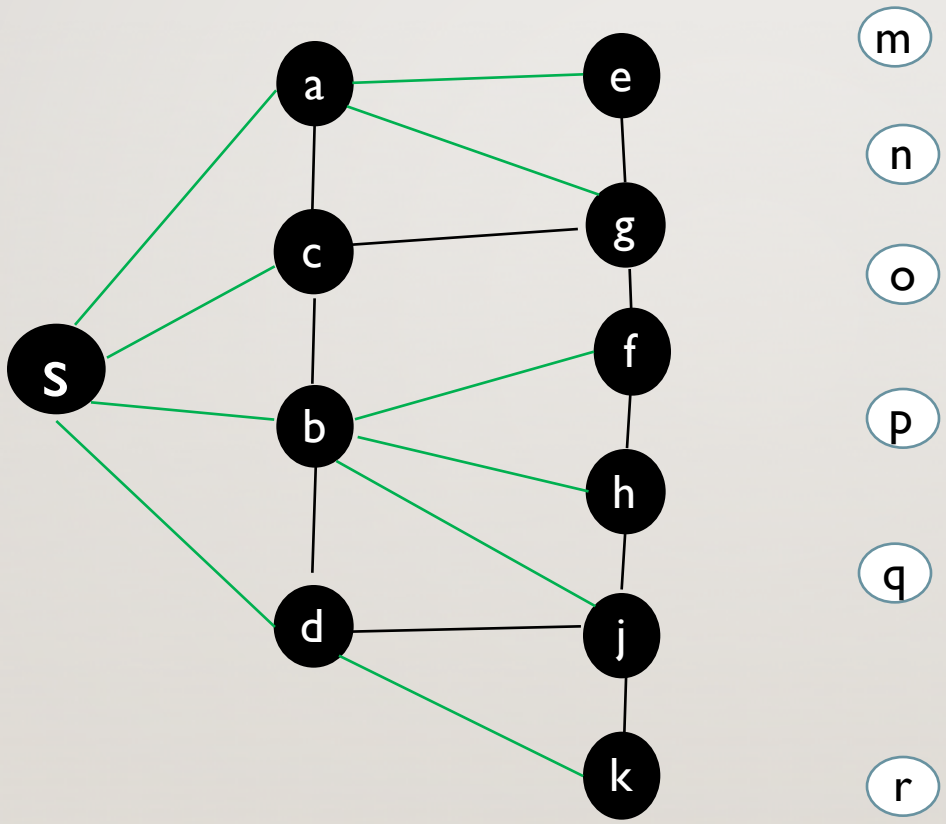

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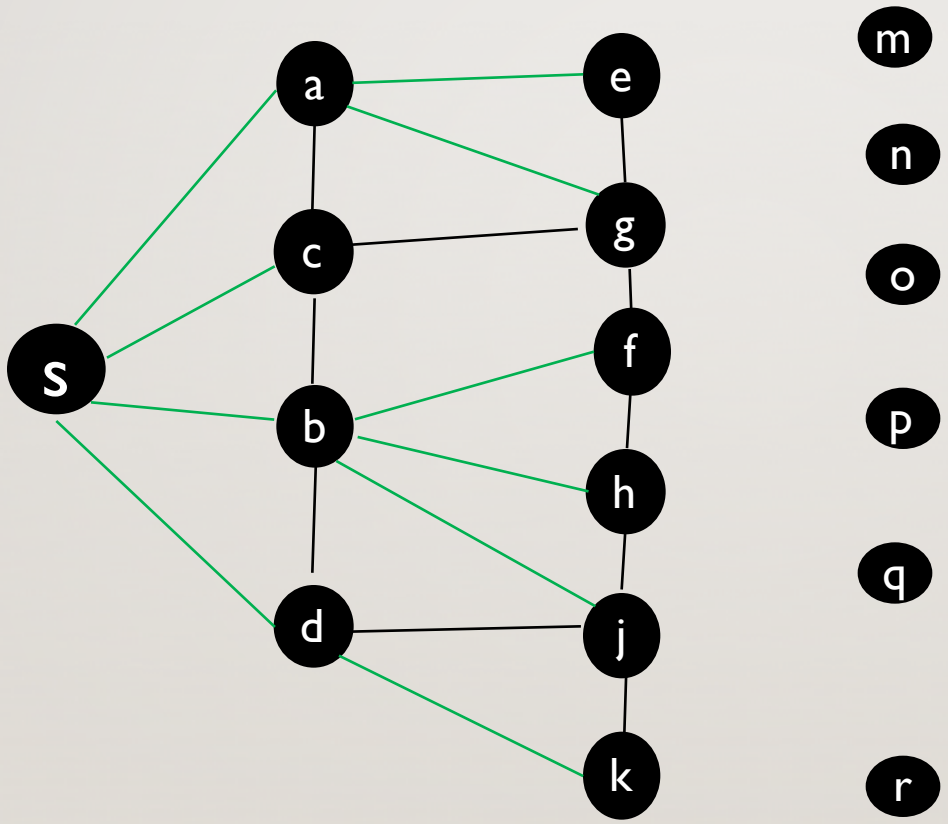

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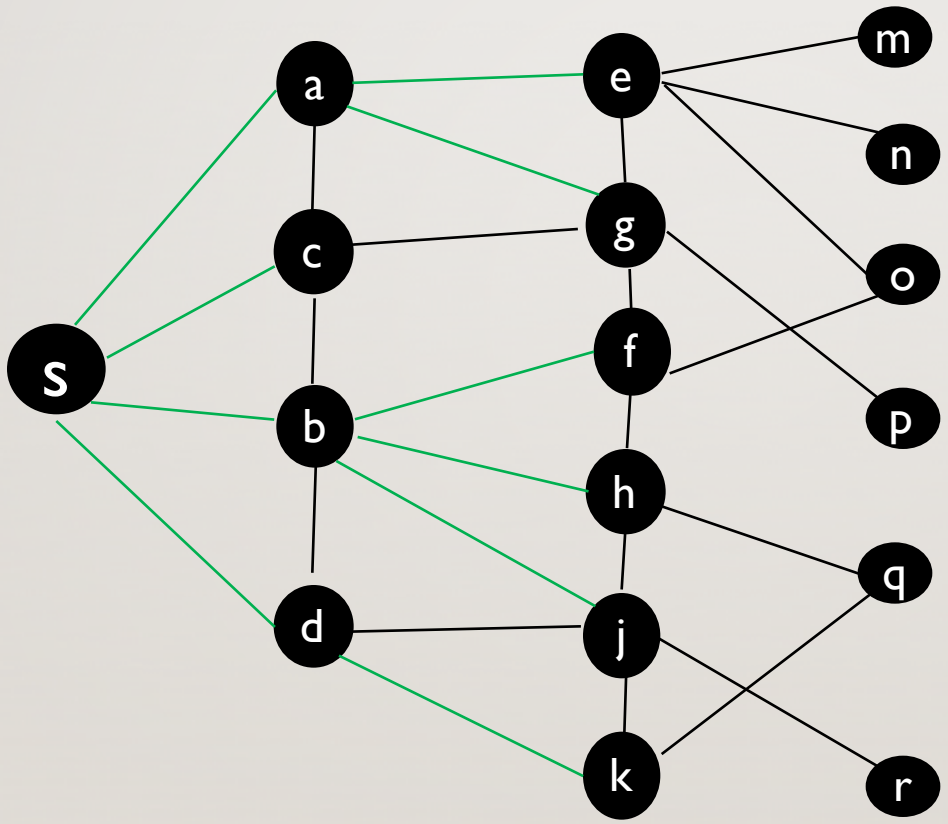

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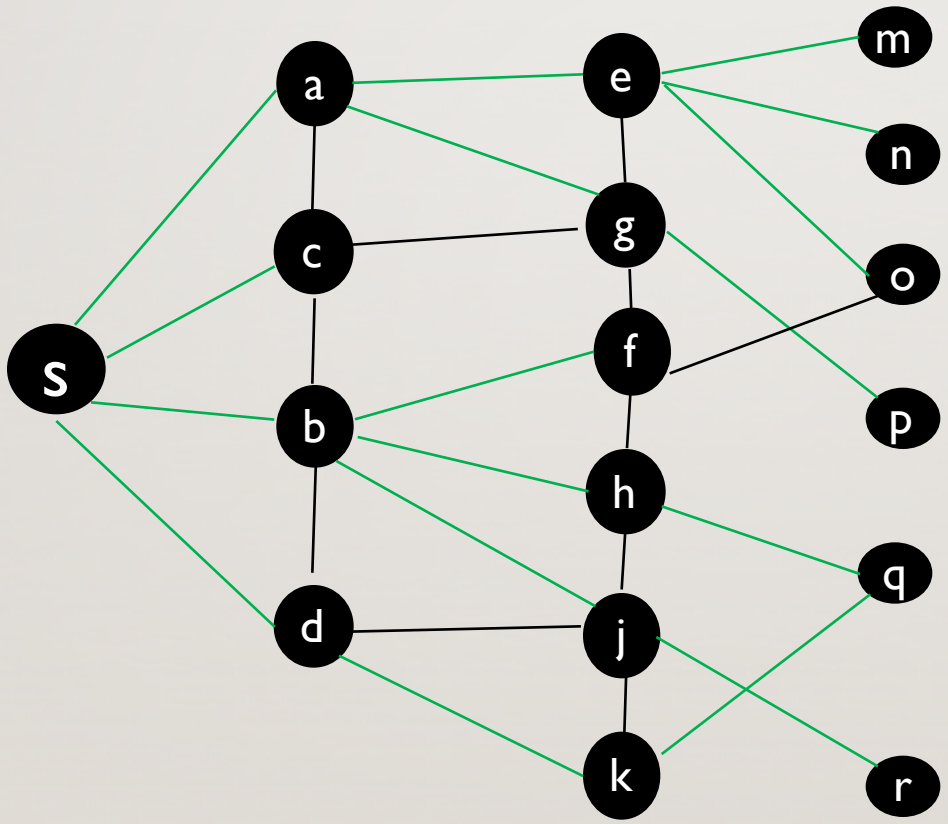

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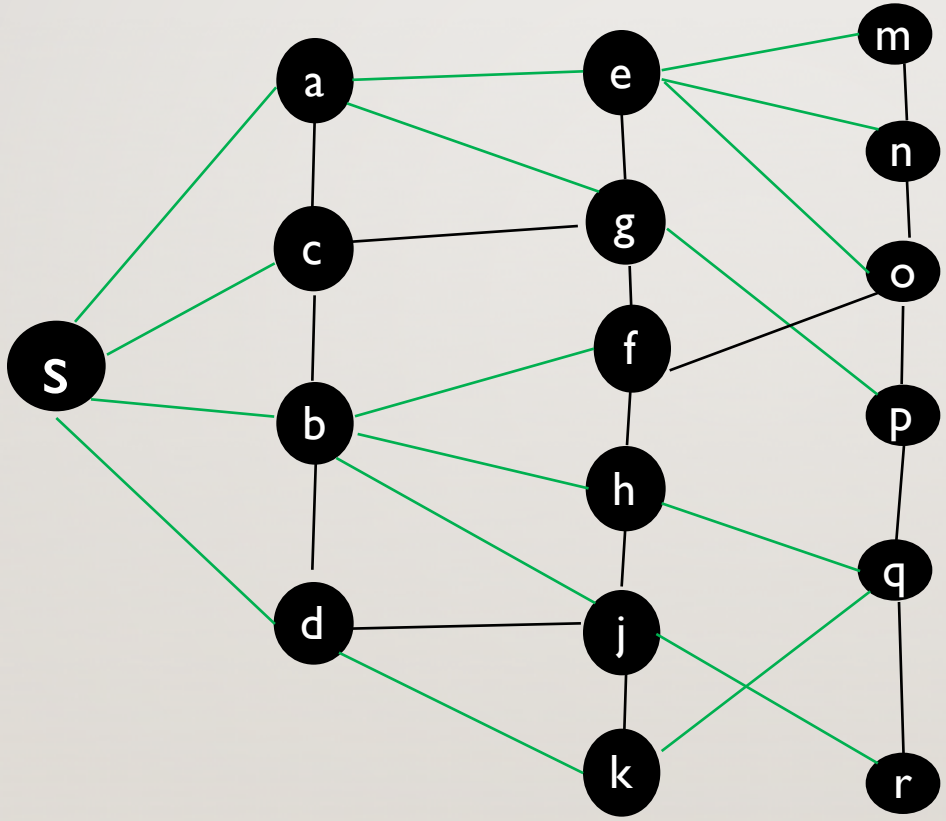
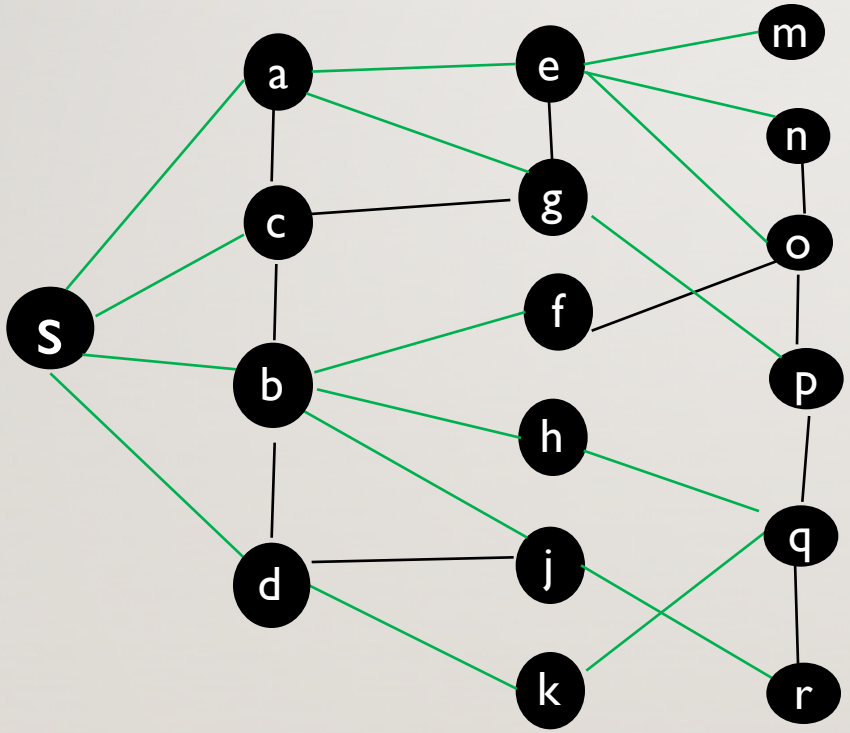


RECAP



- Breadth-First Search is a traversal, it constructs a visiting order of different vertices in the original graph. Following the visiting order, it visits all vertices in the graph one by one
- One product of the Search strategy is the visited edges, in green color. The other edges in the graph is ignored in the traversal.
- The vertices is divided into different equivalent classes; level 1; one move from vertex S. Level 2; two moves from S. Level 3; three movers from S.
- The edges among equivalent classes are ignored in the traversal. Edges between the neighbor level classes might be ignored depending on the visit status.
- It forms a tree connecting all vertices in the graph

BREADTH-FIRST SEARCH ALGORITHM

- We used the natural language to describe the Breadth-First Search Strategy
- How can we describe them in a rigorous algorithm that can be implemented in computer?
- Next, we will use a shortest path searching example to illustrate the rigorous algorithm and then we will work together to write down the implementation code using Psedeo code.
- So far, any questions not clear?